

A Multi-scale Mathematical Model of Growth and Coalescence of Bubbles Beneath the Anode in an Aluminum Reduction Cell



MEIJIA SUN, BAOKUAN LI, and LINMIN LI

Modeling of bubble shapes is challenging because of the wide range of length scales in aluminum reduction cells. A 3D multi-scale mathematical model was developed to understand the nucleation, growth and coalescence of bubbles beneath the anode and to investigate the transition of bubbles from the micro- to macro level. The motion of micro-bubbles is examined using the discrete bubble model (DBM) within a Lagrangian reference. An algorithm for the transition from discrete micro-bubbles to large bubbles, which are fully resolved by the volume of fluid (VOF) approach, is achieved using a user-defined function. The two-way coupling between discrete bubbles and continuous fluids is achieved by inter-phase momentum exchange. The model involves three kinds of bubble coalescence: micro-bubble coalescence is taken into account by the DBM; large bubbles swallowing up micro-bubbles are solved by the discrete-continuum transition model; the coalescence between large bubbles is handled by the VOF method. Numerical results show that the coverage and thickness of bubbles agree well with the experimental data in the literature. Bubble thickness stops increasing when the bubble elongates with the anode bottom, which is 4.0 to 4.5 mm. Meanwhile, two asymmetrical thick heads can be observed under the anode. Bubble release frequency increases with increasing current density.

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I. INTRODUCTION

IN the modern aluminum reduction cell (as shown in Figure 1), the direct current passes through the anode, molten electrolyte (bath) and metal pad to the nearest cathode collector bar. In this process, the electrochemical reaction occurs as the alumina is fed to and dissolved in the molten electrolyte layer at a high temperature of approximately 1223 K. Tiny bubbles are generated under the anode and then grow because of mass transport and coalescence. After that, the larger bubbles leave the nucleation site in a slow and creeping motion. Finally, they move toward the anode edge and release from the anode bottom. The bubbles have both positive and negative effects, such as:

- (a) Improving the mixing and dissolution rates of the alumina in the bath.
- (b) Promoting the heat transfer between the bath and sidewall.
- (c) Reducing the contact area between the bath and anode, reducing the electrochemical reaction rate.
- (d) Increasing the electrical resistance in the anode-cathode distance (ACD) resulting in the increase of voltage drop.
- (e) Increasing of the bath perturbation due to bubble detachment from the anode bottom, resulting in MHD instability.

To take full advantage of the bubble benefits and minimize its negative effects, a better understanding of bubble dynamics in the bath is necessary.

The bubble motion is difficult to measure by ordinary means because of the harsh operating conditions with high temperatures in the aluminum reduction cell. Many physical experiments^[1–9] have been done to investigate the bubble dynamic in the air-water system. Fortin *et al.*^[1] used a full-scale water model of the aluminum reduction cell to study the bubble behaviors. The results indicated that an increase in current density increases the bubble size, bubble thickness and bubble velocity. A higher electrolyte velocity decreases the bubble size and the gas coverage, while the bubble velocity and release

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frequency increase. Vekony *et al.*^[2] developed a real-size air-water electrolysis cell model with two inclined anodes to study the bubble layer under the anode. It was found that the maximum height of bubble is up to 2 cm because of the presence of Fortin bubbles. Simonsen and Einarsrud^[3] studied the bubble coalescence and movement under an inclined surface in a water model of a laboratory-scale electrolysis. The surface tension is increased by adding NaCl. The bubble size in the NaCl solutions is larger than that in the water. Moreover, the NaCl solution appears to be close to a self-organized state. Alam *et al.*^[4] selected the CuSO₄ solution as the electrolyte to achieve the electrolytic reaction and then investigated the bubble formation, coalescence and movement under the horizontal anode surface. Perron *et al.*^[5,6] investigated the influence of surface inclination and liquid properties on the single bubble shape and bubble motion. They pointed out that the terminal velocity of the bubble increases with the bubble volume and surface inclination angle. Kiss *et al.*^[7] developed an air-water model to analyze the mechanisms of bubble departure and bubble detachment under downward-facing solids. It was found that the interplay among different forces (surface forces, buoyancy, flow forces and liquid inertia forces) causes a complex shape evolution for bubbles during different periods. Das *et al.*^[8,9] studied the effect of surface tension on bubble nucleation, formation and onset of sliding based on the air-water experimental model and numerical simulation. The results indicated that the lower surface tension decreases the bubble thickness and the bubble holdup time. The detachment and sliding behavior of the bubble are mainly caused by the balancing act of buoyancy, drag force and surface tension force.

Although physical models are effective to study the bubble behaviors and provide some valuable information to improve our understanding of the bubble dynamics, bubble flow in the air-water system cannot be representative of that in the CO₂-cryolite system in a real aluminum reduction cell. Zhao *et al.*^[10] analyzed the difference in bubble behaviors of the two systems based on the CFD-VOF mathematical model. They pointed out that the CO₂ bubble in the cryolite has a smaller thickness, larger aspect ratio and higher sliding velocity than that of an air bubble in water.

Much research^[11–20] was carried out on analyzing the bubble behaviors in the CO₂-cryolite system, which is closer to the real industrial environment and operational parameter. Xue and Oye^[11] conducted a transparent aluminum electrolysis experiment to study the bubble dynamics (generation, coalescence, growth and detachment) coupled with the cell voltage oscillation. The small voltage fluctuation was found to be associated with bubble coalescence and growth, while a larger amplitude was coupled with the bubble detachment. Cassayre *et al.*^[12] studied the bubble size and bubble layer with oxygen-evolving and graphite anodes. It was concluded that the escaping bubble diameter at an oxygen-evolution anode is much smaller than that at a graphite anode. Good wetting by the electrolyte leads to the formation of tiny bubbles. Zhao *et al.*^[13,14] built a laboratory-scale transparent aluminum reduction cell to

study the anodic bubble behaviors. The results confirmed that the wettability dominates the difference in bubble dynamics between air-water and CO₂-cryolite systems. Various types of bubble coalescence during the bubble growth were also shown.

Although these small-scale, high-temperature electrolysis cells are useful to study the bubble behaviors, their anodes are much smaller than industrial ones because of the high cost of the experimental devices.

At present, numerical modeling is becoming an alternative tool to study bubble behaviors in aluminum reduction cells. Some numerical models^[21–30] have been developed to predict the bubble behaviors. Kiss^[21] developed a mathematical model to describe the individual bubble growth based on the finite element method (FEM). They found that the bubble growth rate is mainly affected by the diffusion resistance and the gas storage capacity inside the anode solid element. Perron *et al.*^[22,23] analyzed the effects of bubble shape and wetting film thickness on relative resistance using FEM. The results indicated that the bubble shape slightly affects the electrical resistance. Some researchers^[24,25] adopted the volume of fluid (VOF) method to capture the bubble-bath interface. This method is applicable to describe the larger bubble, which is flattened under the anode surface. However, for tiny bubbles, it requires a large number of grids to obtain the accurate solutions of the small bubble-bath interfaces, resulting in increasing computation burden. Others^[26–30] used the Euler-Euler method and population balance model (PBM)^[31–34] to simulate the bubble distribution. Only Einarsrud^[31] *et al.* developed a two-dimensional multi-scale model to investigate the sub-grid bubbles and resolved bubbles using the PBM method coupled with the VOF method. However, the bubble size distribution was not discussed. The Lagrangian discrete bubble model (DBM) was developed to track the bubble trajectory.^[35] With the limitation of this model, the bubbles are always spherical and the deformation of larger bubbles cannot be observed. Both the concept of coupling a Lagrange model to VOF and coupling a sub-grid PBM model to VOF are intended to handle the coexistent small dispersed and large deformed bubbles with saving of computing resources. In the PBM model, bubble distribution is always calculated based on the volume average method. However, the single bubble trajectory cannot be tracked and the bubble interface cannot be resolved. In the DPM model, although the bubble interface cannot be resolved, the bubble trajectory can be traced, and a direct physical interpretation is also given. This multi-scale nature of bubbles is a challenge to numerical simulation. A pathway to comprehensive modeling still needs to be found.

The main objective of the present work is to propose a new transition model from micro-bubbles to large ones to handle the coexistent small discrete and large deformed bubbles. A multi-scale mathematical model using the DBM and VOF methods is developed to track the micro-bubbles and capture the large bubble-bath interface, respectively. The growth of discrete bubbles is accounted for based on the Faraday law. The collision and coalescence of discrete bubbles are simulated using

the DBM model. The large deformed bubbles swallowing up the small discrete bubbles, *i.e.*, discrete-continuum transition, is described by coupling the DBM and VOF. The coalescence between large deformed bubbles is also fully resolved by the VOF method. To validate the model, the simulation results are compared with previous experimental data.

II. MODEL FORMULATION

In the aluminum reduction cell, the bubble behaviors as well as the complex transport phenomena cannot be observed and measured in the harsh operation environment. Hence, numerical simulation, as an effective method, should be considered to understand the complicated process. To maintain a reasonable computational time, some assumptions are put forward:

1. The bubble and bath are incompressible Newtonian fluids.
2. The electrochemical reaction between the anode and bath is disregarded.
3. The temperature field is not considered.
4. The anode surface roughness and gas cavities are not taken into account. The bubble nucleation is uniform at the anode surface.

A. Discrete Bubble Model for Micro-bubbles

The DBM^[36–38] is an effective approach to tracking bubbles that are small, spherical and dispersed considering coalescence. The motion of bubbles in the flow field can be described in the Lagrangian coordinates by integrating the force balance on bubbles. According to the Newton's law of motion, the following equations describe the motions of a spherical bubble immersed in fluid considering the buoyancy, drag force, virtual mass force and pressure gradient force. Moreover, bubble mass is calculated based on bubble growth model, which is further introduced in the next part.

$$m_b \frac{d\vec{u}_b}{dt} = \vec{F}_b + \vec{F}_d + \vec{F}_{vir} + \vec{F}_p \quad [1]$$

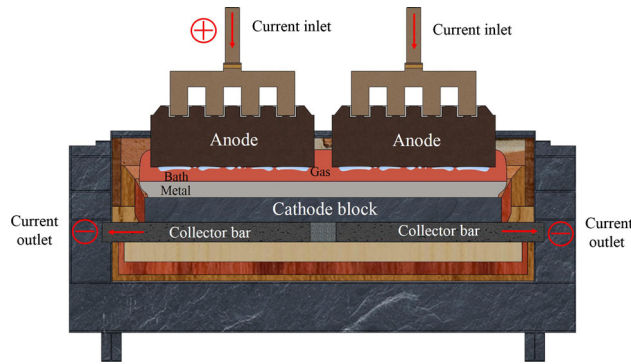


Fig. 1—Schematic diagram of an aluminum reduction cell.

$$\frac{dm_b}{dt} = \frac{\rho_b RTA}{4NFP_2} I \quad [2]$$

where $\vec{F}_b$ is the result of buoyancy and gravity, which is written by:

$$\vec{F}_b = m_b \frac{\vec{g}(\rho_b - \rho)}{\rho_b} \quad [3]$$

$\vec{F}_d$ is the drag force, which can be expressed as:

$$\vec{F}_d = \frac{3}{4} \rho V_b \frac{C_D}{d_b} |\vec{u} - \vec{u}_b| (\vec{u} - \vec{u}_b) \quad [4]$$

where C_D is the drag coefficient that can be described by the spherical drag law.^[39]

The virtual mass force $\vec{F}_{vir}$ plays an important role in bubbles in liquid^[40,41] and is written as:

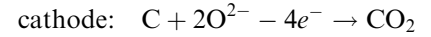
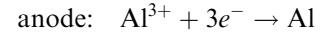
$$\vec{F}_{vir} = \frac{1}{2} \rho V_b \frac{d}{dt} (\vec{u} - \vec{u}_b) \quad [5]$$

$\vec{F}_p$ is the pressure gradient force, which is given as:

$$\vec{F}_p = \rho V_b \vec{u}_b \nabla \vec{u} \quad [6]$$

B. Micro-bubble Growth

In the typical Hall–Heroult cell, the electrochemical reaction occurs at 1223 K when the current flows from the anode to the cathode. The alumina (Al_2O_3) is dissolved in the molten bath, and the simplified electrode reactions at the anode and cathode are:



Hence, 1 mol of Al generates 3/4 mol of CO_2 . The amount of carbon dioxide n_{CO_2} is calculated based on the Faraday law,

$$n_{\text{CO}_2} = \frac{It}{3F} \times \frac{3}{4} = \frac{It}{4F} \quad [7]$$

where I is the current, t is the time, and F is the Faraday constant of 96,485 C/mol (Figure 2).

According to the ideal gas equation, the volume of CO_2 can be defined as:

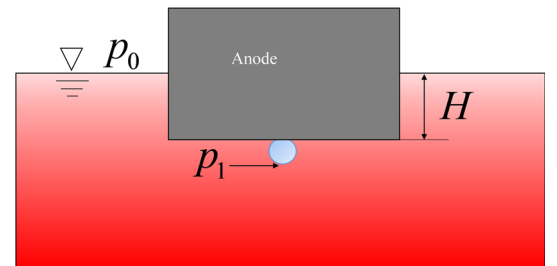


Fig. 2—Schematic diagram of bubble pressure.

$$V_{\text{CO}_2} = \frac{IRT}{4Fp_1} t \quad [8]$$

$$p_1 = p_0 + \rho_1 g(H + 2r) \quad [9]$$

where p_0 is atmospheric pressure, p_1 is bubble pressure in the bath, ρ_1 is bath density, H is the anode immersion depth, and r is the bubble radius. The gas production at the anode bottom is:

$$q = V_{\text{CO}_2} A = \frac{IRTA}{4Fp_1} t \quad [10]$$

Some experimental observations^[11,14] indicate that small bubbles are generated at the anode bottom as semi or full sphere forms. The bubble radius can be written as:

$$r = \sqrt[3]{\frac{3}{16} \frac{RTA}{N\pi Fp_1} It} \quad [11]$$

$$r \propto \sqrt[3]{It} \quad [12]$$

where R is the molar gas constant, T is the temperature, A is the anode bottom area, and N is the total number of bubbles.

In this bubble growth model, the bubble volume increases with increasing current and time. However, in the real condition, the large bubble deforms because of the unbalance forces and cannot grow in the spherical pattern. This model is only suitable for the initial growth of bubble nucleation. In this section, bubble nucleation sites are not explored, which is beyond the scope of our study and needs to be addressed in future work.

C. Micro-bubble Coalescence

The bubbles can collide and coalesce from small bubbles to large ones. According to the O'Rourke's algorithm,^[36,38,42] for description the bubbles with larger radii are called collectors and those with smaller radii are called small bubbles. When the small bubble passes within a flat circular center around the collector of area $\pi(r_1 + r_2)^2$, a collision will take place. The probability of collision is defined as:

$$P = \frac{\pi(r_1 + r_2)^2 \vec{u}_{\text{rel}} \Delta t}{V_{\text{cell}}} \quad [13]$$

where r_1 is the radius of the larger bubble, r_2 is the radius of the smaller bubble, $\vec{u}_{\text{rel}}$ is the relative velocity between two bubbles, Δt is the time step, and V_{cell} is the cell volume.

The actual probability distribution of N_c collisions follows a Poisson distribution:

$$P(N_c) = e^{-\bar{N}} \frac{\bar{N}^{N_c}}{N_c!} \quad [14]$$

with the mean value of expected number collisions $\bar{N} = N_1 P = \frac{N_1 \pi (r_1 + r_2)^2 \vec{u}_{\text{rel}} \Delta t}{V_{\text{cell}}}$. N_c is the number of collisions between a larger bubble and other bubbles; N_1 is the number of the smaller bubbles.

To determine whether the two bubbles collide or rebound, the first random number x , range (0, 1), is assigned only if $x > P$ (0) so that a collision can take place.^[43] Then, the second random number, y ($y \in (0, 1)$), is assigned and the actual collision parameter b is defined as:

$$b = (r_1 + r_2) \sqrt{y} \quad [15]$$

If $b < b_{\text{crit}}$, where b_{crit} is the critical offset, then the result of every collision is coalescence. The critical offset is expressed as:

$$b_{\text{crit}} = (r_1 + r_2) \sqrt{\min\left(1.0, \frac{2.4f}{\text{We}}\right)} \quad [16]$$

where f is a function of the relative radius of the larger bubble and the smaller one and We is the collisional Weber number. They are defined as:

$$f\left(\frac{r_1}{r_2}\right) = \left(\frac{r_1}{r_2}\right)^3 - 2.4\left(\frac{r_1}{r_2}\right)^2 + 2.7\left(\frac{r_1}{r_2}\right) \quad [17]$$

$$\text{We} = \frac{\rho_b \vec{u}_{\text{rel}} \bar{d}}{\sigma} \quad [18]$$

where $\bar{d}$ is the arithmetic mean diameter of two bubbles.

After coalescence, the larger bubble swallows the smaller one to generate a new larger bubble. The mass and velocity of the new bubble are calculated based on the conservation law. O'Rourke^[42] derived the following expression for the new mass and velocity:

$$m'_1 = m_1 + m_2 \quad [19]$$

$$m'_1 \vec{u}'_b = m_1 \vec{u}_{b1} + m_2 \vec{u}_{b2} \quad [20]$$

If $b \geq b_{\text{crit}}$, the rebound occurs between the two bubbles. The bubble mass is not changed while the bubble velocity can be updated.

$$\vec{u}'_{b1} = \frac{m_1 \vec{u}_{b1} + m_2 \vec{u}_{b2}}{m_1 + m_2} + \frac{m_2 (\vec{u}_{b1} - \vec{u}_{b2})}{m_1 + m_2} \left(\frac{b - b_{\text{crit}}}{r_1 + r_2 - b_{\text{crit}}} \right) \quad [21]$$

$$\vec{u}'_{b2} = \frac{m_1 \vec{u}_{b1} + m_2 \vec{u}_{b2}}{m_1 + m_2} + \frac{m_1 (\vec{u}_{b1} - \vec{u}_{b2})}{m_1 + m_2} \left(\frac{b - b_{\text{crit}}}{r_1 + r_2 - b_{\text{crit}}} \right) \quad [22]$$

where m_1 is the mass of larger bubble, m_2 is the mass of smaller bubble, $\vec{u}_{b1}$ is the velocity of larger bubble, $\vec{u}_{b2}$ is the velocity of smaller bubble, $\vec{u}'_b$ is the new bubble velocity after coalescence, $\vec{u}'_{b1}$ is the velocity of larger bubble after rebound, and $\vec{u}'_{b2}$ is the velocity of smaller bubble after rebound.

D. VOF Method for Large Deformed Bubbles

The surface of the large deformed bubble is tracked by the VOF method^[44] by solving a continuity equation for the volume fraction:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \vec{u}) = 0 \quad [23]$$

The flow field is modeled by a continuity equation and a single set of momentum equations.^[45]

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = S_1 + S_2 \quad [24]$$

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u} \vec{u}) = & -\nabla p + \nabla \cdot [\mu (\nabla \vec{u} + \nabla \vec{u}^T)] \\ & + \vec{F}_{me} + \vec{F}_s \end{aligned} \quad [25]$$

where S_1 and S_2 are source terms of dispersed micro-bubbles transforming to the resolved ones, which are described in the next part. ρ is the mixture density and μ is the mixture viscosity, which are defined as:

$$\rho = \alpha \rho_l + (1 - \alpha) \rho_g \quad [26]$$

$$\mu = \alpha \mu_l + (1 - \alpha) \mu_g \quad [27]$$

In the present work, the two-way coupling is achieved *via* the sink term $\vec{F}_{me}$, which represents the momentum exchange between the discrete bubble (Lagrangian) and continuous liquid (Eulerian), such as drag force, virtual mass force and pressure gradient force.

$$\begin{aligned} \vec{F}_{me} = & \left(\frac{3}{4} \rho V_b \frac{C_D}{d_b} |\vec{u} - \vec{u}_b| (\vec{u} - \vec{u}_b) \right. \\ & \left. + \frac{1}{2} \rho V_b \frac{d}{dt} (\vec{u} - \vec{u}_b) + \rho V_b \vec{u}_b \nabla \vec{u} \right) \Delta t \end{aligned} \quad [28]$$

$\vec{F}_s$ is the surface tension using the continuum-surface-force (CSF) model developed by Brackbill,^[46] which can be expressed as:

$$\vec{F}_s = \sigma_{ij} \kappa \nabla \alpha \quad [29]$$

where σ_{ij} is the surface tension coefficient and κ is the curvature calculated as:

$$\kappa = -\nabla \cdot (\nabla \alpha / |\nabla \alpha|) \quad [30]$$

E. Discrete-Continuum Transition

The bubble behaviors are complicated in the aluminum reduction cell. In the early stage of the electrochemical reaction, the bubbles are quite small and spherical. Then, the discrete micro-bubbles can grow to larger bubbles, which are deformable because of the mass transfer and coalescence. The discrete bubbles are tracked by the DBM within a Lagrangian reference, while the large bubbles are resolved by the VOF

method. Hence, it is necessary to achieve the discrete-continuum transition to reveal bubbles varying in a wide range of sizes.

The present work considers the two types of transition from discrete micro-bubbles to resolved ones. One is that, when a discrete bubble grows to a large bubble, which can be resolved, we resolve it using the VOF method and stop tracing it in the Lagrangian reference. The other is that when a discrete bubble attaches the resolved bubble and they are estimated to have aggregated with each other, a source is added to the resolved bubble and the discrete bubble is removed. The schematic diagram for the first kind of transition is given in Figure 3. In the numerical model, a source equaling the mass of the removed bubble is added to the current location once when the discrete bubble is removed and then handled with the VOF method using a user-defined function (UDF). The source is calculated as:

$$S_1 = \sum_{i=1}^n \rho_g V_{b_i} (i = 1, 2, 3, \dots, n) \quad [31]$$

$$u_{g_i} = u_{b_i} \quad [32]$$

where ρ_g is the resolved bubble density.

For the second one, the micro-bubble remains close to stationary under the anode; when the large bubble sweeps alongside the anode, the large bubble will engulf the micro-bubble, as sketched in Figures 4(a) and (b). It is assumed that when the discrete bubble touches the large bubble interface where $0.5 \leq \alpha_g \leq 1$, it will be absorbed, as seen in Figure 4(c). The mass of the discrete bubble is added to the continuity equation as a source term, and the velocity of fluid in the current cell can also be updated as:

$$S_2 = \sum_{i=1}^n \rho_g V_{b_i} \quad (i = 1, 2, 3, \dots, n) \quad [33]$$

$$u_{g'} = u_{g_i} + u_{b_i} \frac{V_{b_i}}{V_{cell}} \quad [34]$$

Figure 5 describes a flow chart of the combined DBM-VOF algorithm for bubble behaviors. The flow chart shows the two-way coupling *via* the momentum exchange sink term $\vec{F}_{me}$ between the two solvers for discrete bubbles and resolved fields. All equations are marched with the same time step. The dispersed bubbles will always be tracked with the flow time of the continuous flow solver. In this simulation, it is divided into two modules including different functions of each operational module: one is the discrete bubble coalescence; the other is the discrete-continuum transition. The initial discrete bubbles are small and spherical, and they are tracked using the DBM solver. Then, the discrete bubbles grow and coalesce to larger ones. If $V_b \geq V_{cell}$ or $0.5 \leq \alpha_g \leq 1$, which satisfies the criteria for the transitions, the discrete bubbles will transform to the resolved ones.

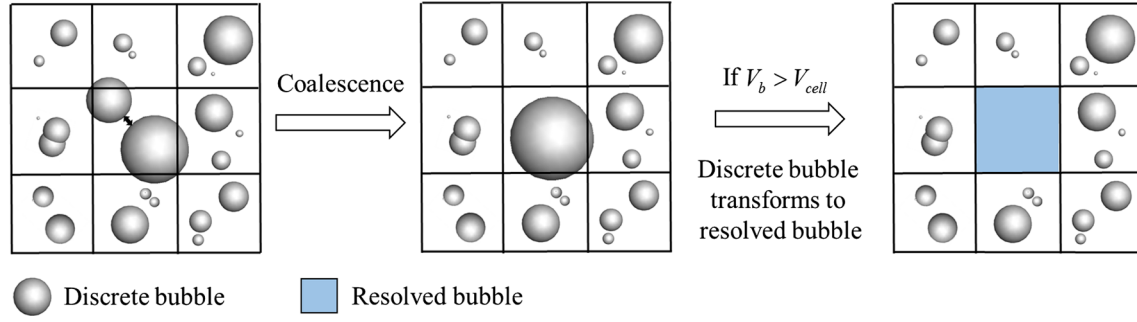


Fig. 3—Schematic transitions from a discrete bubble to resolved bubble with $V_b \geq V_{cell}$.

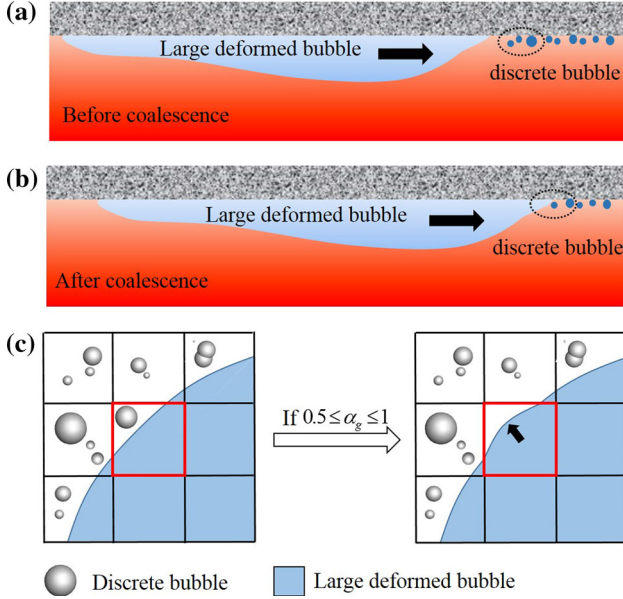


Fig. 4—Sketch of a large deformed bubble swallowing a discrete bubble with $0.5 \leq \alpha_g \leq 1$: (a) before coalescence; (b) after coalescence; (c) transition process.

III. NUMERICAL DETAILS

In the present work, the CFD software Fluent 15.0 with UDF is used to solve perform the simulations based on the finite volume method. The computation domain is discretized with a hexahedron mesh as shown in Figure 6. The geometrical parameters and material properties are listed in Table I.

To access the grid dependence of the numerical results, we conduct three cases including 362,000, 198,000 and 134,000 cells, respectively, as shown in Figure 7. In view of the comparative nature of the different grids, the deviation of the simulated result of the average velocity between the first and second meshes is about 2.4 pct, while it is approximately 3.9 pct between the second and third mesh. Considering the calculation time and accuracy, a total number of 198,000 cells is retained for this study to predict the multiscale flow field. The mesh size is about 0.8 mm.

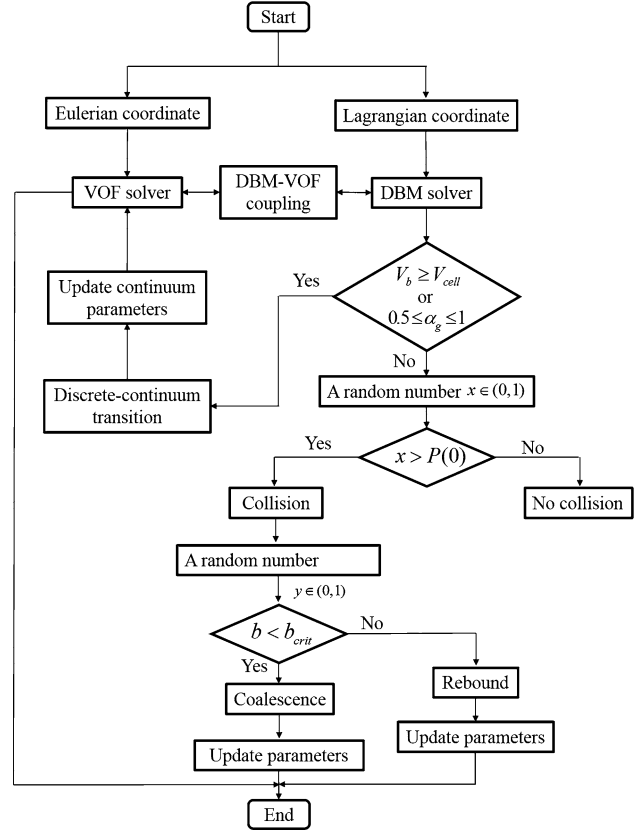


Fig. 5—Flow chart of the combined DBM-VOF algorithm.

In the calculation, the velocity inlet boundary of dispersed micro-bubbles is adopted at the anode bottom. The top surface of the cell is set to be the outflow boundary, and a zero shear stress is employed. The later wall of the anode and the later wall of the cell and cell bottom are supposed to be non-slipping wall boundaries. The initial bubble diameter is set to 0.5 mm, which is the minimal bubble size observed in some laboratory experiments.^[11,13,14] The bubble growth is based on the Eq. [11]. When bubbles grow or coalesce to a specified volume that is larger than the cell volume, then the discrete micro-bubbles will transform to resolved ones using the UDF module.

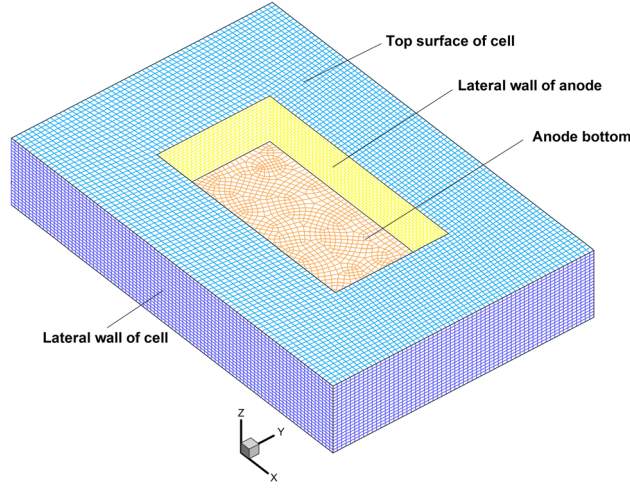


Fig. 6—Mesh for simulation.

Table I. Geometrical and Physical Parameters

Parameter	Value
Current density (A/m^2)	9000
Anode bottom (mm)	22×50
Cell (mm)	$90 \times 60 \times 20$
Physical properties of bath	
Density (kg/m^3)	2130
Viscosity [$kg/(m \cdot s)$]	0.002513
Physical properties of gas	
Density (kg/m^3)	0.398
Viscosity [$kg/(m \cdot s)$]	0.00005
Surface tension (N/m)	0.12

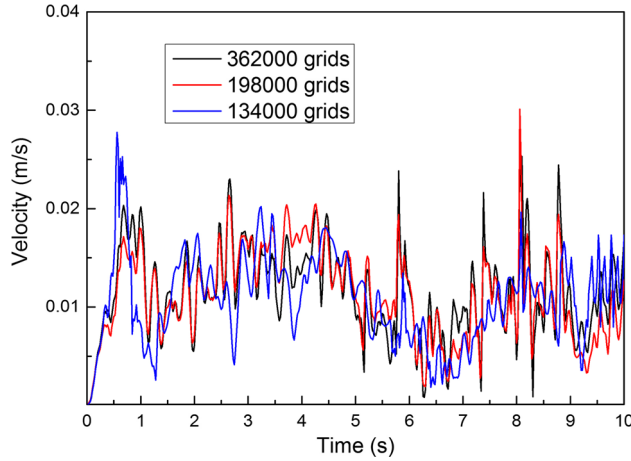


Fig. 7—Comparison of the effect of the number of meshes on point (30, 0, 7 mm) velocity.

Given the characteristic of fluid flow in the experiment referenced by Zhao *et al.*,^[13] it does not reveal any turbulent behavior. Hence, the transient laminar flow model is adopted to calculate the bubble-bath two-phase flow with a low Reynolds number. The phase coupled SIMPLE scheme is used for the pressure-velocity coupling of the gas-bath flow and the transient scheme for volume fraction equations, which will be

set to a bounded second-order scheme. The flow time step is set to be 0.0001 seconds to meet the convergence criteria (10^{-6}).

IV. RESULTS AND DISCUSSION

A. Model Validation

Figure 8 shows comparisons of bubble coverage and bubble thickness between experimental data and the simulation result considering the influence of micro- and macro-bubbles. We tracked micro-bubbles' trajectories and resolved macro-bubbles' surfaces at every time step to calculate the average bubble thickness and coverage. Good agreement was found. This indicated that the model developed in the present work is suitable for predicting the bubble coverage and thickness in aluminum reduction cells. Also, the fluctuation frequencies agree well with each other.

B. Discrete Bubble Coalescence

In the initial bubble growth stage, the discrete micro-bubbles are almost spherical, so the present model assumes that the micro-bubbles are spherical and their coalescence is modeled. The continuous process of micro-bubble coalescence in this simulation is presented in Figure 9. It shows that two micro-bubbles first collide with each other when moving below the anode and then coalesce into a larger bubble. The volume of the larger bubble is equal to the sum of that of the two small bubbles. Meanwhile, the larger bubble velocity is updated based on Eq. [20].

C. Discrete-Continuum Transition

Figure 10 shows the discrete-continuum transition process. In Figure 10(a), the bubbles are tracked by DBM in the Lagrangian scheme, and the bubble sizes and locations are updated at every time step. One criterion based on the bubble volume is set to achieve the discrete-continuum transition. If $V_b \geq V_{cell}$, the transition will be accomplished where the bubble center is located, as presented in Figure 10(b). Then the bubble interface is resolved by the mesh within the Eulerian coordinate.

Figure 11 shows the phenomenon of a large resolved bubble swallowing up the discrete micro-bubble. As mentioned in Section II-E, another criterion based on the gas volume fraction is proposed. If the discrete bubble reaches the position where $0.5 \leq \alpha_g \leq 1$, it will be absorbed; in addition, the mass and velocity of the resolved bubble will be updated.

D. Large Deformed Bubble Coalescence

Figure 12 displays large bubble coalescence in the present simulation and shows that the micro-bubbles coexist with the large ones. After micro-bubbles transform to large ones, the bubbles are deformed by the buoyancy, surface tension and pressure gradient forces.

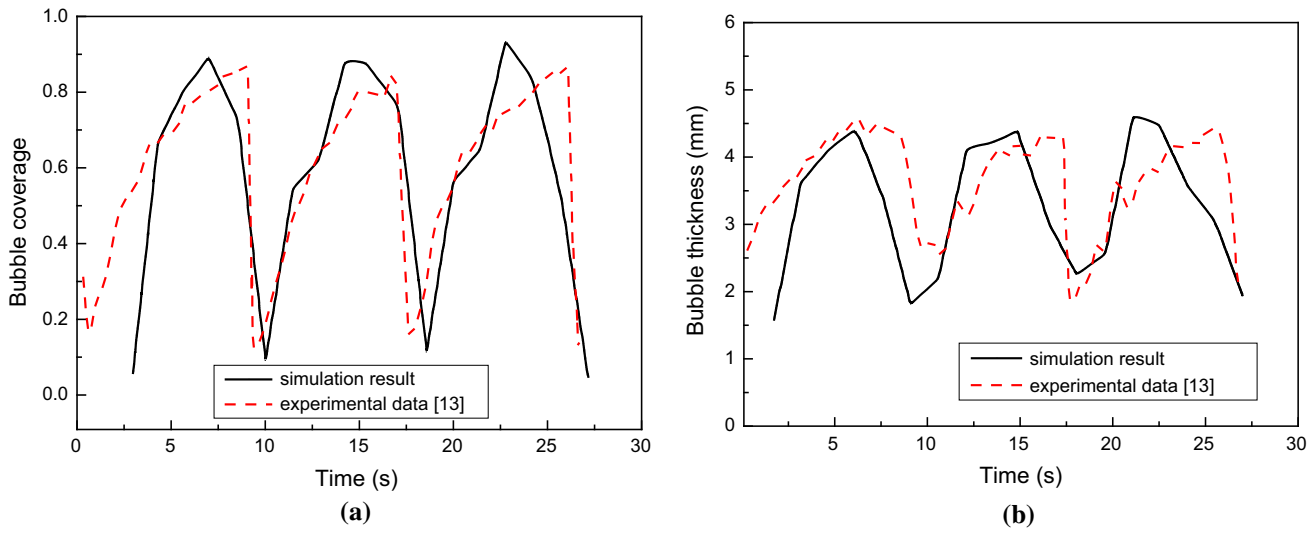


Fig. 8—Comparisons of (a) bubble coverage and (b) bubble thickness between experimental data and simulation results with current density of 9000 A/m².

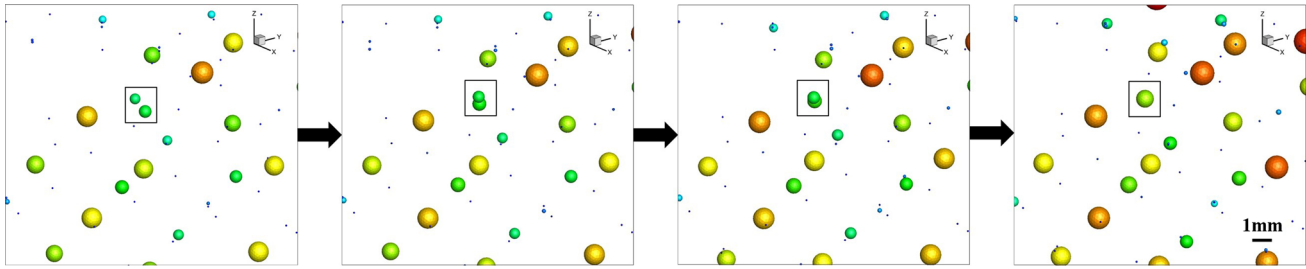


Fig. 9—Discrete bubble coalescence process.

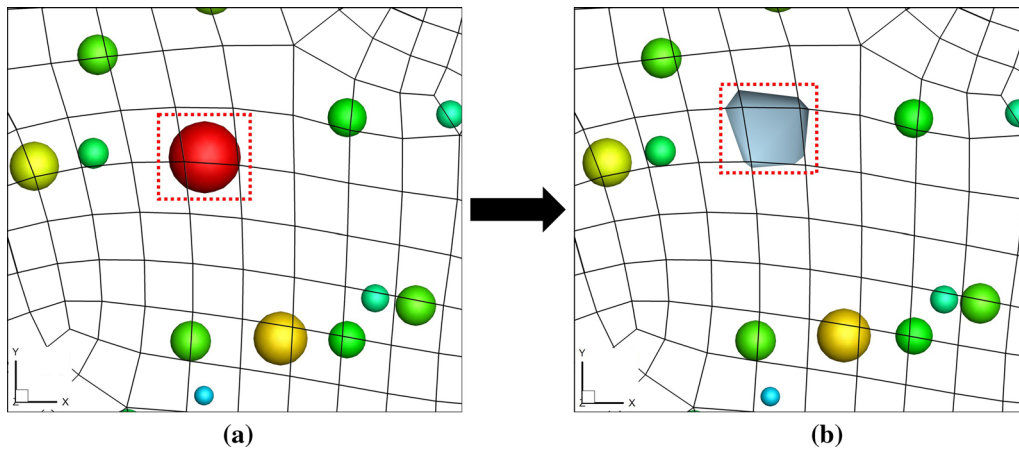


Fig. 10—Discrete-continuum transition if $V_b \geq V_{cell}$: (a) before transforming; (b) after transforming.

The large bubble can absorb the nearby discrete ones so that its volume can be increased. Zhao *et al.*^[13,14] distinguished six types of large bubble coalescence in the small high-temperature electrolytic cell, which are shown in Figures 12(a) through (f): two small-size bubbles coalesce into a larger bubble (S-S coalescence); a medium-size bubble swallows a small-size bubble (M-S coalescence); two medium-size bubbles coalesce

into a large-size bubble (M-M coalescence); a large-size bubble and a medium-size bubble coalesce into a larger bubble (L-L coalescence); a large-size bubble and a small-size bubble coalesce into a larger bubble (L-S coalescence); two large-size bubbles coalesce into a larger bubble (L-L coalescence). It is indicated that the present model can well reveal the bubble coalescence process.

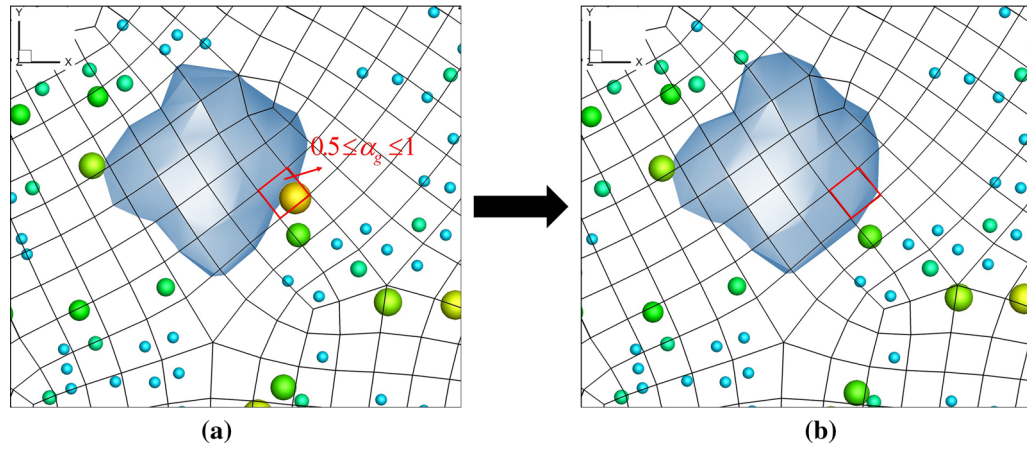


Fig. 11—Large deformed bubble swallowing up the discrete micro-bubble if $0.5 \leq \alpha_g \leq 1$: (a) before swallowing up; (b) after swallowing up.

E. Bubble Coverage and Bubble Release Frequency

Figure 13 shows simulation results of gas coverage and bubble release frequency with various current densities from 0 to 60 seconds. A sawtooth-like wave pattern can be observed, which reflects the bubble birth, growth and detachment. The bubble release frequency with the largest amplitude is coupled with bubble detachment from the anode bottom. It is demonstrated that the bubble release frequency increases from 0.035 to 0.196 with increasing current density from 1000 A/m^2 to $10,000 \text{ A/m}^2$. High current accelerates the electrochemical reaction rate, and then the bubble formation and growth rates can be increased. Hence, the critical state of bubble release is achieved quickly. This will shorten the bubble residence time to prevent larger bubble formation, leading to a smaller bubble size. Meanwhile, a careful observation reveals that the maximum gas coverage is reduced slightly with increasing current density, which will help increase the contact area between the anode bottom and molten bath; thus, the cell electrical conductivity can be improved. Figure 14 shows that the bubble release frequency increases with current density. In Xue^[11] and Zhao's^[13] experiments, the anode size of the former is smaller than that of the latter, which decreases the bubble residence time in the cell and increases the bubble release frequency. The present simulation is performed at room temperature based on Zhao's experimental model. We assume that the anode wettability does not change with temperature, and the contact angle of the bath and anode is set to 90. However, in their experiments, the contact angle of the bath and anode is about 100 to 120. A smaller contact angle makes a better wettability,^[13] which decreases the distance of rising bubbles from vertical surfaces and increases the bubble release frequency.

F. Bubble Thickness

The evolution of bubble thickness is presented in Figure 15. The bubbles transform from spherical to elliptical shape as they grow further. This is mainly caused by the buoyancy force, which is perpendicular to the anode bottom, resulting in flattening of bubbles. The

bubble thickness increases as bubble coalescence takes place, while the number of bubbles decreases. Then, the bubble starts to spread over the anode bottom surface when it reaches a specified thickness and stops growing. The maximum bubble thickness is about 4.2 mm, consistent with some researchers' observations.^[8,13,14,18,47,48] The Fortin bubble is observed when the bubble volume is large enough, as shown in Figure 15(h). The bubble elongates in the two opposite directions during spreading from the anode center to anode edge under the horizontal surface. Some researchers^[5,6] found one thick head and thin tail with an inclined anode surface. However, in the present work, two thick heads can be observed, and the thin tail cannot be clearly found because of the horizontal anode bottom. Although the cell is a symmetric system, the bubble motion is random at the anode bottom. Various-size bubbles coalesce with each other, resulting in an asymmetric bubble distribution. Moreover, the velocities of bubbles are different after collisions.

Figure 16 gives the evolution of maximum and average bubble thicknesses along with current density. The maximum bubble thickness reaches to 4.55 mm when the current density is 7500 A/m^2 and then decreases with increasing current density. Moreover, the average bubble thickness always decreases with increasing current density considering the influence of micro- and macro-bubbles. According to our simulation, the micro-bubble has little effect on the whole bubble thickness. The result illustrates that a high current affects bubble thickness by increasing the bubble release frequency. The high current may weaken the influence of bubble dynamics on the molten bath by decreasing the bubble thickness, which shortens the ACD to save energy.

V. CONCLUSIONS

In the present work, a 3D DBM-VOF model with a discrete-continuum transition is developed to predict the bubble-bath two-phase flow in aluminum reduction cell. To investigate the multi-scale bubble behaviors, the

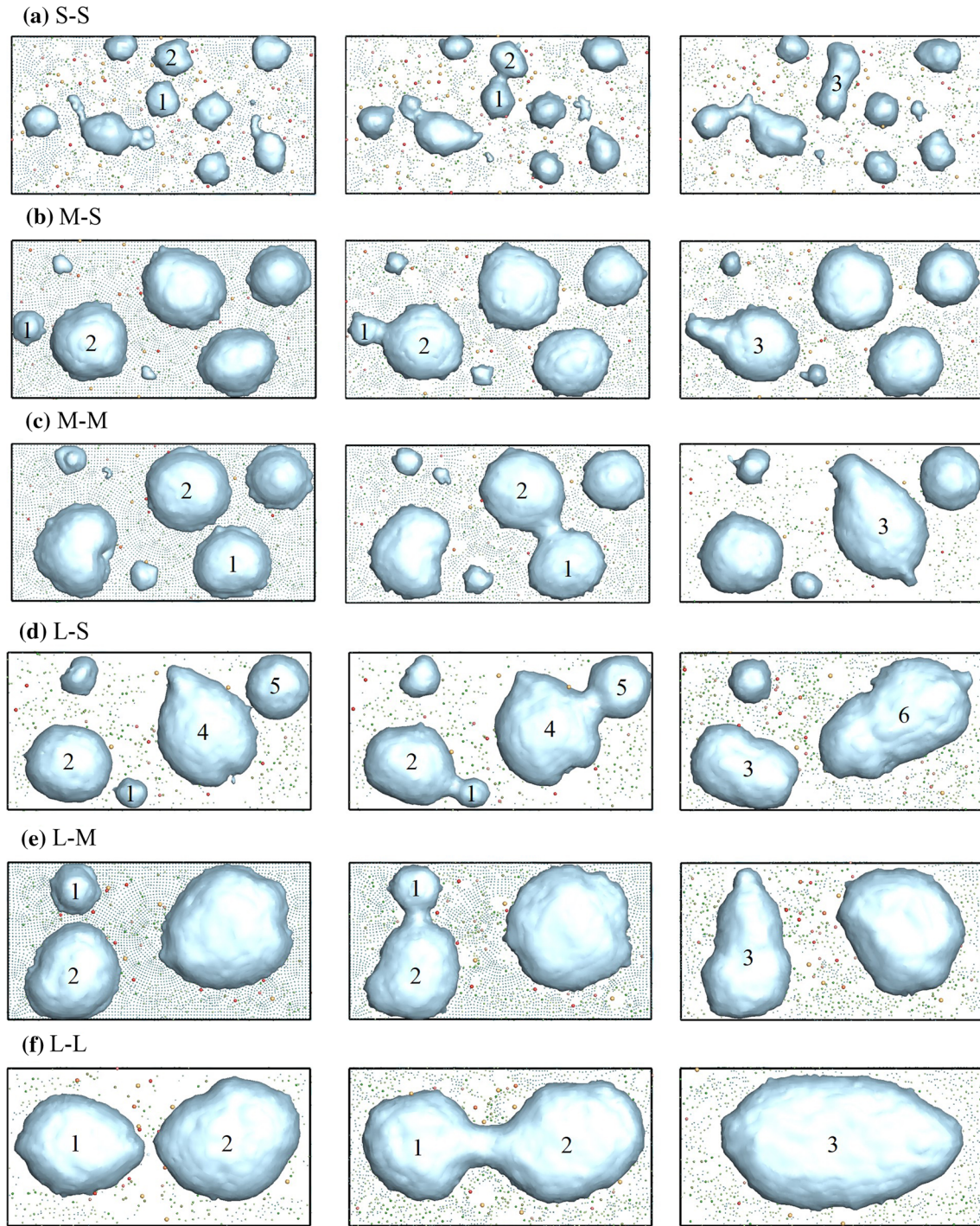


Fig. 12—Large bubble coalescence process. (a) S-S; (b) M-S; (c) M-M; (d) L-S; (e) L-M; (f) L-L.

discrete micro-bubbles are modeled using the DBM, while the large bubbles are fully resolved from macroscopic view by the VOF method. A new transition from a discrete to large bubble is achieved by coupling with UDF modules. Several conclusions can be drawn as follows:

1. Three kinds of bubble coalescence are achieved in the present model: the coalescence between discrete micro-bubbles is observed using DBM; a large de-

formed bubble swallowing up the discrete bubble is described by coupling DBM and VOF; the coalescence between large resolved bubbles is also revealed with VOF.

2. The bubble thickness increases as the bubble coalescence takes place, while the number of bubbles decreases. Then, the bubble starts to spread over the anode bottom surface when it reaches a specified thickness and stops growing. Two thick heads can be

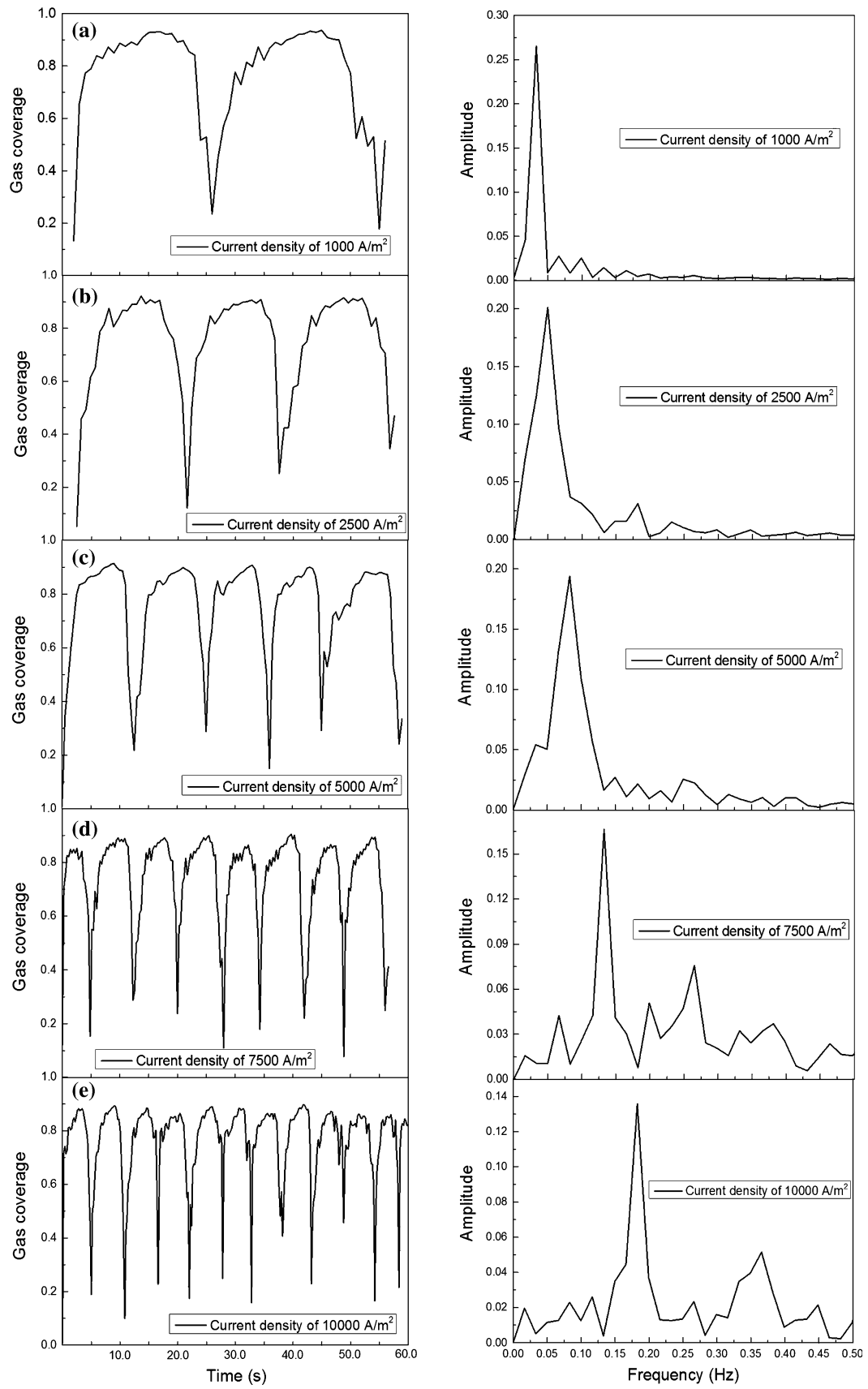


Fig. 13—Gas coverage and bubble release frequency with various current densities: (a) 1000 A/m²; (b) 2500 A/m²; (c) 5000 A/m²; (d) 7500 A/m²; (e) 10,000 A/m².

observed, while the thin tail cannot be clearly found because of the horizontal anode bottom.

3. The bubble release frequency increases and the gas coverage decreases with increasing current density leading to a smaller release bubble size; thus, the anode-bath contact area can be increased to improve

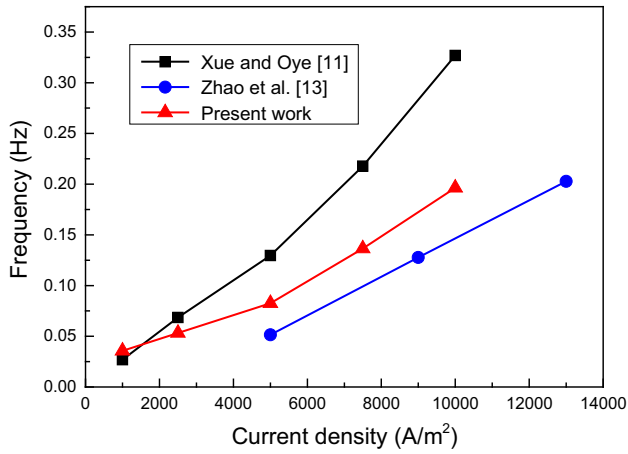


Fig. 14—Evolution of bubble release frequency with current density.

the cell electrical conductivity. Bubble thickness is slightly affected by current density, which is 4.0 to 4.5 mm in the present work. The simulation results of gas coverage and gas thickness are compared with experimental data from Zhao *et al.*,^[13,14] and good agreement is obtained.

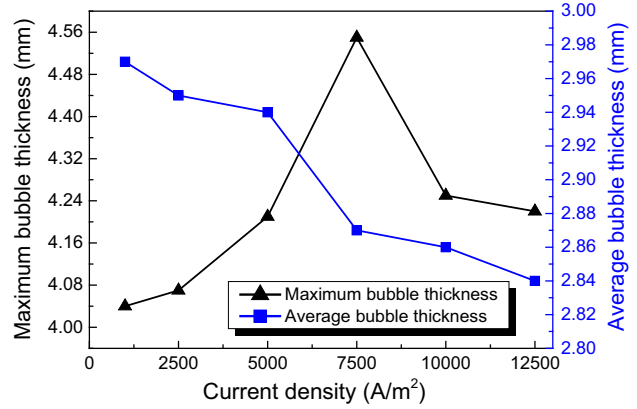


Fig. 16—Evolution of maximum and average bubble thicknesses along with current density.

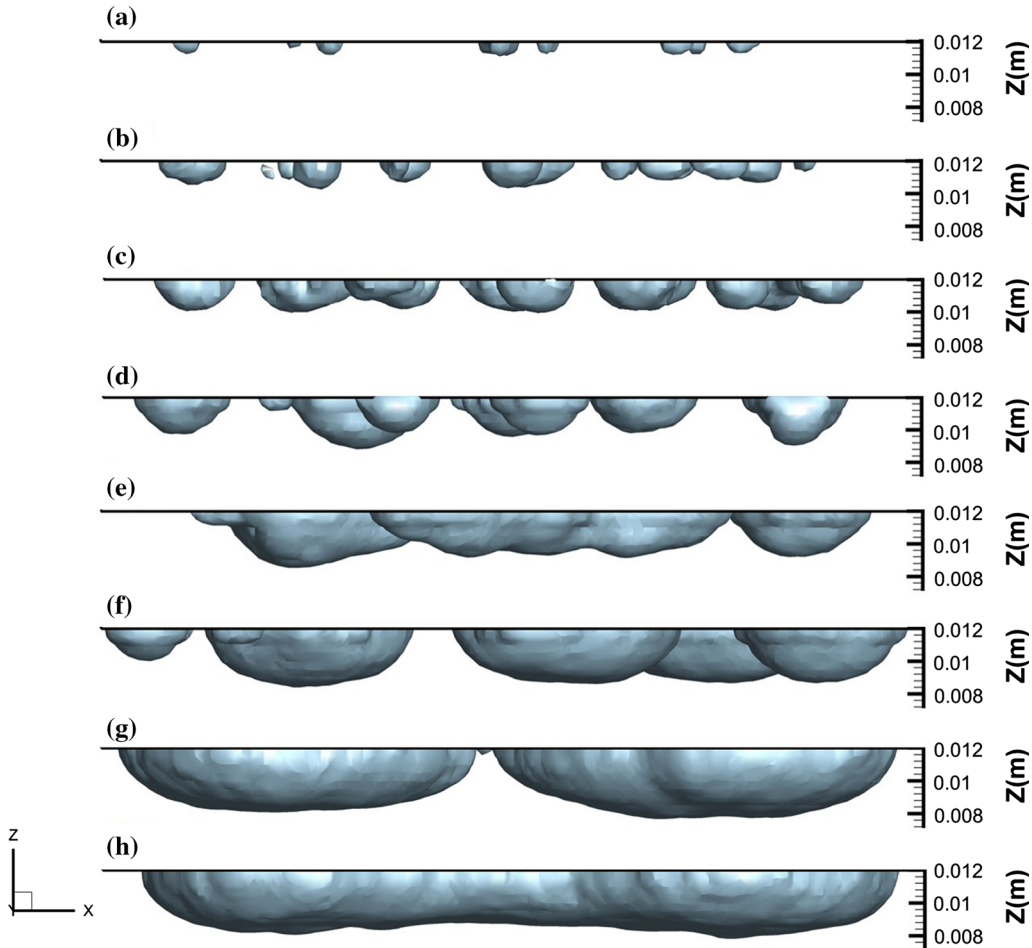


Fig. 15—Bubble thickness. (a) 2.5 s; (b) 3.1 s; (c) 3.7 s; (d) 4.3 s; (e) 4.9 s; (f) 5.5 s; (g) 6.1 s; (h) 6.7 s.

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NOMENCLATURE

m_b	Bubble mass
m'_1	New bubble mass after coalescence
m_1	Larger bubble mass
m_2	Smaller bubble mass
$\vec{u}_b$	Bubble velocity
$\vec{u}'_b$	New bubble velocity after coalescence
$\vec{u}_{b1}$	Larger bubble velocity
$\vec{u}_{b2}$	Smaller bubble velocity
$\vec{u}'_{b1}$	Larger bubble velocity after rebound
$\vec{u}'_{b2}$	Smaller bubble velocity after rebound
u_g	Resolved bubble velocity
$\vec{u}$	Bath velocity
$\vec{u}_{rel}$	Relative velocity
$\vec{F}_b$	The resultant force of buoyancy and gravity
$\vec{F}_d$	Drag force
$\vec{F}_{vir}$	Virtual mass force
$\vec{F}_p$	Pressure gradient force
$\vec{F}_{me}$	Momentum exchange
$\vec{F}_s$	Surface tension
V_b	Bubble volume
V_{cell}	Cell volume
d_b	Bubble diameter
$\bar{d}$	Arithmetic mean diameter of two bubbles
r	Bubble radius
r_1	Larger bubble radius
r_2	Smaller bubble radius
C_D	Drag coefficient
I	Current
F	Faraday constant
R	Molar gas constant
p_1	Bubble pressure in the bath
H	Anode immersion depth
Δt	Time step
P	Collision probability
N	Total number of bubbles
$\bar{N}$	Mean collision number
N_1	Number of smaller bubbles
N_c	Number of collisions
x, y	Random numbers (0, 1)
b	Actual collision parameter
b_{crit}	Critical offset
We	Collisional Weber number
S_1, S_2	Mass source

GREEK SYMBOLS

ρ	Mixture density
ρ_b	Bubble density
ρ_g	Resolved bubble density
ρ_1	Bath density
μ	Mixture viscosity

μ_l	Bath viscosity
μ_g	Resolved bubble viscosity
σ_{ij}	Surface tension coefficient
κ	Curvature
α	Volume fraction
α_g	Gas volume fraction

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